Notation and Setup for Design-Based Causal Inference*

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This guide collects, in one place, the notation and formal setup shared by the companion guides on (i) the unbiasedness of the Difference-in-Means estimator for both the Sample Average Treatment Effect (SATE) and the Population Average Treatment Effect (PATE), (ii) the variance of the estimator for the SATE and the estimation of that variance, and (iii) the variance of the estimator for the PATE and the estimation of that variance. Each of those guides gives just enough of the setup for its own proofs. This guide develops that setup in full. A reader who is already comfortable with the potential outcomes framework for randomized experiments can safely skim this guide.

1 Units, assignments, and the assignment mechanism

Let the index $i \in \{1, \ldots, n\}$ run over n units in a finite sample, $\mathcal{S}_n$, where $n \geq 4$. Of these n units, the design assigns $n_T \geq 2$ to the treatment condition and $n_C \geq 2$ to the control condition, where $n_T + n_C = n$. The bounds $n_T \geq 2$ and $n_C \geq 2$ (and hence $n \geq 4$) play no role in the unbiasedness of the estimator or in the variance formula itself; they ensure that the conservative (Neyman) estimator of the variance (developed in a companion guide) is well defined. That estimator computes a separate sample variance within each treatment arm, dividing by $n_T - 1$ for the treated units and by $n_C - 1$ for the control units. A sample variance requires at least two observations, so the estimator needs at least two units in each arm.

Let the binary indicator variable $Z_i \in \{0, 1\}$ record whether unit i receives treatment ($Z_i = 1$) or control ($Z_i = 0$), and collect these indicators in the random assignment vector $\boldsymbol{Z} = \begin{bmatrix} Z_1, \ldots, Z_n \end{bmatrix}^\top$. The set of all logically possible assignment vectors is $\{0, 1\}^n$.

Every assignment vector determines how many units it sends to treatment: Write $n_T(\boldsymbol{z}) \coloneqq \sum_{i=1}^{n} z_i$ for this count of treated units, and $n_C(\boldsymbol{z}) \coloneqq n - n_T(\boldsymbol{z})$ for the count of control units. Evaluated

[*] This is a live document that is subject to updating at any time.

at the random assignment vector, the counting function for treated units yields $n_T(\boldsymbol{Z})$, itself a random variable and a function of the random assignment vector in exactly the way the potential outcomes below are functions of their assignment argument. The assignment mechanism places positive probability only on a subset of the 2^n logically possible vectors. Under *complete random assignment* the design fixes the count of treated units, restricting the support to

$$\Omega = \left\{ \boldsymbol{z} \in \{0,1\}^n : n_T(\boldsymbol{z}) = n_T \right\},$$

where n_T is a constant that the experimenter chooses.

On Ω the random count of treated units, $n_T(\boldsymbol{Z})$, equals the constant n_T with probability one, so I write the plain constant, just as z_i denotes the realized value of Z_i. Under complete random assignment, $\boldsymbol{Z}$ is distributed uniformly on Ω, and the number of elements in Ω, denoted $|\Omega|$,[1] is $\binom{n}{n_T}$. By contrast, n independent Bernoulli assignments would place positive probability on all 2^n vectors. The definition of complete random assignment thus has two separable parts, and the guides use both: The fixed count of treated units determines the support, and uniformity is a further assumption about the distribution the mechanism places on that support.

I distinguish three objects that are easily conflated. The *assignment space* $\{0,1\}^n$ is the set of all conceivable assignments. The *support* $\Omega \subseteq \{0,1\}^n$ is the (typically smaller) set of assignments the mechanism can actually produce. And the *randomization distribution* is the probability distribution the mechanism places on Ω (uniform, under complete random assignment). All of the randomness in what follows comes from this distribution.

One convention holds throughout the guides: Ω always denotes the support of $\boldsymbol{Z}$ under the assignment mechanism in force, and when an argument moves between mechanisms I condition explicitly rather than change the symbol. Holding the count of treated units fixed and collecting every assignment vector that achieves that count also builds the support of a stratified design: A blocked or matched study fixes a count of treated units within each stratum, and the support of that study's assignment vector collects the assignment vectors that hit all of those counts at once.

The same framework accommodates *simple random assignment*, under which the mechanism assigns each of the n units independently, with a common assignment probability $p \in (0,1)$ for every unit. Two features change. First, no design constraint pins down the count of treated units, so $n_T(\boldsymbol{Z})$ is a nondegenerate random variable. Second, I exclude the two degenerate assignment vectors (all treated and all control), under which one group is empty and the Difference-in-Means estimator is undefined; the support is therefore $\Omega = \{\boldsymbol{z} \in \{0,1\}^n : 0 < n_T(\boldsymbol{z}) < n\}$, which has $2^n - 2$ elements. The exclusion of those two vectors changes the distribution, so I state the distribution explicitly:

[1]For an arbitrary set $\mathcal{W}$, let $|\mathcal{W}|$ denote the cardinality of (i.e., the number of elements in) the set $\mathcal{W}$.

$\boldsymbol{Z}$ follows the distribution of n independent Bernoulli draws with success probability p, *conditioned on* the event $\{0 < n_T(\boldsymbol{Z}) < n\}$, and the conditioning breaks the independence of the coordinates.

Note where the probabilities live: Simple random assignment defines them at the level of individual units, and the distribution of the whole vector follows as a consequence. An assignment $\boldsymbol{z}$ in the support has probability proportional to $p^{n_T(\boldsymbol{z})}(1-p)^{n-n_T(\boldsymbol{z})}$, which depends on $\boldsymbol{z}$ only through its count of treated units and is uniform on the support only when $p = \frac{1}{2}$. Complete random assignment runs in the opposite direction: The design defines the distribution at the level of the whole vector, uniform on Ω, and the probabilities for individual units follow as consequences, which the unbiasedness guide derives.

Even though the design does not fix $n_T(\boldsymbol{Z})$, we can fix the count of treated units by conditioning on its realized value: The randomization distribution conditional on the event $\{n_T(\boldsymbol{Z}) = n_T\}$ is exactly the complete random assignment distribution above, whatever the value of p. Analyzing a coin-flip experiment as if it had been a lottery with a fixed number of winning tickets is exactly this conditioning (Rosenbaum, 2017, pp. 289–290, fn. 15), and Pashley et al. (2021) give general conditions under which such conditional analyses are valid. The unbiasedness guide proves the reduction of simple random assignment to complete random assignment. Hence results that I derive under complete random assignment extend to simple random assignment after conditioning on the realized number of treated units.

2 The potential outcomes schedule and the Stable Unit Treatment Value Assumption

I adopt the terminology of Freedman (2009) and later Gerber and Green (2012), and define a potential outcomes schedule as a vector-valued function $\boldsymbol{y} : \{0,1\}^n \to \mathbb{R}^n$ that maps each possible assignment vector $\boldsymbol{z} \in \{0,1\}^n$ to an n-dimensional vector of real-valued outcomes. The schedule lists how each of the n units would respond to every assignment $\boldsymbol{z}$ that the experiment could in principle produce. The schedule is a *fixed* feature of the units; randomness enters only through which assignment the mechanism selects. I move in numbered steps from this general object to the familiar pair of potential outcomes: for each unit, one fixed outcome under treatment and another under control. I will use the same four-step format, general object followed by assumption invoked followed by licensed simplification followed by realization, whenever an assumption changes what the notation must express.

Step 1 (general object). Unit i's outcome under assignment $\boldsymbol{z}$ is the ith coordinate of the schedule's value, written $y_i(\boldsymbol{z})$. Composed with the random assignment, $y_i(\boldsymbol{Z})$ is a random variable that could in principle take as many as 2^n distinct values, one for each assignment vector.

Step 2 (assumption invoked). Under the Stable Unit Treatment Value Assumption (SUTVA)[2] (Cox, 1958; Rubin, 1980, 1986), unit i's outcome depends on the assignment vector only through unit i's own coordinate: For any $\boldsymbol{z}, \boldsymbol{z}'$ with $z_i = z_i'$, $y_i(\boldsymbol{z}) = y_i(\boldsymbol{z}')$.

Step 3 (licensed simplification). By Step 2, the function $\boldsymbol{z} \mapsto y_i(\boldsymbol{z})$ depends on $\boldsymbol{z}$ only through z_i, so $y_i(1)$ is well defined as the common value of $y_i(\boldsymbol{z})$ across all $\boldsymbol{z}$ with $z_i = 1$, and $y_i(0)$ as the common value across all $\boldsymbol{z}$ with $z_i = 0$. SUTVA thus collapses 2^n numbers per unit to two fixed numbers per unit. The individual causal effect for unit i on the additive scale is $\tau_i \coloneqq y_i(1) - y_i(0)$. The vectors $\boldsymbol{y}(1)$ and $\boldsymbol{y}(0)$ collect the treatment and control potential outcomes for all n units, and $\boldsymbol{\tau}$ collects the n individual effects; all three vectors are fixed, finite population quantities.

Step 4 (realization). The observable outcome for unit $i \in \{1, \dots, n\}$ is

$$Y_i \coloneqq y_i(Z_i) = Z_i y_i(1) + (1 - Z_i)\, y_i(0),$$

which is random only through Z_i; under the realized assignment $\boldsymbol{Z} = \boldsymbol{z}$, the observed outcome is the number $y_i(z_i)$. Potential outcomes are fixed, the assignment is random, and observable outcomes are random only because a random assignment selects which fixed potential outcome we see.

3 Estimands and the Difference-in-Means estimator

Write $\bar{y}(1) \coloneqq n^{-1} \sum_{i=1}^{n} y_i(1)$ and $\bar{y}(0) \coloneqq n^{-1} \sum_{i=1}^{n} y_i(0)$ for the finite population means of the treatment and control potential outcomes; both are fixed quantities. The target of interest is the Sample Average Treatment Effect (SATE),

$$\tau_{\text{SATE}} \coloneqq n^{-1} \sum_{i=1}^{n} \tau_i = n^{-1} \sum_{i=1}^{n} \big(y_i(1) - y_i(0)\big) = \bar{y}(1) - \bar{y}(0),$$

which is likewise a fixed (though unknown) quantity. The Difference-in-Means estimator of τ_{SATE} is

$$\hat{\tau} \coloneqq \frac{\sum\limits_{i=1}^{n} Z_i Y_i}{\sum\limits_{i=1}^{n} Z_i} - \frac{\sum\limits_{i=1}^{n} (1 - Z_i)\, Y_i}{\sum\limits_{i=1}^{n} (1 - Z_i)}.$$

[2]SUTVA implies that (1) each unit in the experiment responds only to the condition to which the design assigns that unit and (2) the treatment condition is actually the same treatment for all units assigned to treatment and the control condition is the same for all units assigned to control.

Unlike the estimand τ_{SATE}, the estimator $\hat{\tau}$ is a *random variable*, since $\hat{\tau}$ is a function of the random assignment vector $\boldsymbol{Z}$ and inherits all randomness from $\boldsymbol{Z}$. The denominators are the counts of treated and control units, $n_T(\boldsymbol{Z})$ and $n_C(\boldsymbol{Z})$, which in general vary from one assignment to another. Under complete random assignment the design fixes those two counts at n_T and n_C with probability one, so we can equivalently write $\hat{\tau}$ as

$$\hat{\tau} = \frac{1}{n_T} \sum_{i=1}^{n} Z_i Y_i - \frac{1}{n_C} \sum_{i=1}^{n} (1 - Z_i) \, Y_i.$$

Both displays are ratios, and they differ only in what stands in their denominators: The first divides by the two counts as random variables, the second by the two constants that the design fixes. I use the random-denominator form when the count of treated units may vary across assignments, as under simple random assignment, and the fixed-denominator form when the design fixes that count.

Because the only randomness is the assignment, I write $\mathrm{E}_{\Omega}[\cdot]$ and $\mathrm{Var}_{\Omega}[\cdot]$ for expectations and variances over the distribution of $\boldsymbol{Z}$ on Ω, to emphasize that they pertain to the randomization distribution alone. The subscript names the distribution in force on Ω, not the set itself. In these guides that distribution is the uniform one just described. However, other distributions are possible on the same support, as when a sensitivity analysis considers departures from uniform assignment.

4 Finite population variances of the potential outcomes

The variance results use finite population variances of the potential outcomes, and I follow two conventions for those variances, differing only in the divisor. The first convention uses the divisor $n - 1$:

$$\begin{aligned}
S_n^2(\boldsymbol{y}(1)) &= \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} \left(y_i(1) - \bar{y}(1)\right)^2 \\
S_n^2(\boldsymbol{y}(0)) &= \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} \left(y_i(0) - \bar{y}(0)\right)^2 \\
S_n^2(\boldsymbol{\tau}) &= \left(\frac{1}{n-1}\right) \sum_{i=1}^{n} \left(\tau_i - \tau_{\text{SATE}}\right)^2.
\end{aligned}$$

The second convention uses the divisor n and is the one that Gerber and Green (2012, Equation 3.4) uses:

$$\sigma_n^2(\boldsymbol{y}(1)) = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \left(y_i(1) - \bar{y}(1)\right)^2$$

$$\sigma_n^2(\boldsymbol{y}(0)) = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \big(y_i(0) - \bar{y}(0)\big)^2$$
$$\sigma_n(\boldsymbol{y}(0), \boldsymbol{y}(1)) = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \big(y_i(0) - \bar{y}(0)\big) \big(y_i(1) - \bar{y}(1)\big) \,.$$

The two conventions are related by $\sigma_n^2(\cdot) = \frac{n-1}{n} S_n^2(\cdot)$, and likewise for the covariance term. All of the variances under both conventions are fixed, finite population quantities, and they will be the building blocks of the variance of $\hat{\tau}$. The naming rule deserves one explicit statement: The letter S marks the divisor one less than the size, the letter σ marks the divisor equal to the size itself, the subscript names the index set (n for the sample $\mathcal{S}_n$, N for the superpopulation $\mathcal{P}_N$), and the absence of a subscript marks the infinite superpopulation limit introduced at the end of the next section.

5 Extension to a superpopulation: the Population Average Treatment Effect

For results about the Population Average Treatment Effect (PATE), I embed the finite sample in a superpopulation. Consider a superpopulation $\mathcal{P}_N$ of size $N \geq n \geq 4$, with index $i \in \{1, \ldots, N\}$, and let simple random sampling draw n units from $\mathcal{P}_N$ into the experimental sample $\mathcal{S}_n$, leaving the remaining $N - n$ units unsampled. Among the n sampled units, complete random assignment sends $n_T \geq 2$ to treatment and $n_C \geq 2$ to control, exactly as above. The companion guides develop the superpopulation results under complete random assignment at this second stage; the variance guides close with extensions to simple random assignment.

The binary indicator $R_i \in \{0, 1\}$ records whether unit i enters the sample ($R_i = 1$) or stays out of it ($R_i = 0$), and I collect these indicators in $\boldsymbol{R} = \Big[R_1, \ldots, R_N\Big]^\top$. Write $n(\boldsymbol{r}) \coloneqq \sum_{i=1}^{N} r_i$ for the number of units a sampling vector selects; the support of $\boldsymbol{R}$ is then $\Pi = \{\boldsymbol{r} \in \{0, 1\}^N : n(\boldsymbol{r}) = n\}$, which holds the count of sampled units fixed exactly as Ω holds the count of treated units fixed, now at the sampling stage.

Simple random sampling, in the form the guides use, has a sample size fixed by design: The mechanism selects one of the $\binom{N}{n}$ subsets of exactly n of the N units uniformly at random sampling *without replacement* which is the same as distributing $\boldsymbol{R}$ uniformly on Π. Every draw therefore produces exactly n sampled units, so $n(\boldsymbol{R}) = n$ with probability one, and I write the plain constant n, exactly as I write n_T under complete random assignment. The qualifier *fixed by design* matters because a different mechanism, *Bernoulli sampling*, would flip a coin for each of the N units separately and so make the sample size $n(\boldsymbol{R})$ a nondegenerate random variable, exactly as simple random assignment makes $n_T(\boldsymbol{Z})$ one. Throughout the guides the sample size stays fixed at n, even in the infinite superpopulation model of the PATE variance guide, which draws a fixed number n of units independently.

One symbol Ω serves every sample, and the license for that reuse deserves its own statement. The assignment stage operates on whichever units the sampling stage selects, so in full generality the support of the assignment vector could depend on the realized sample: Write $\Omega_{\boldsymbol{r}}$ for the set of ways to assign the units that $\boldsymbol{r}$ selects. Two features of the design make the single symbol sufficient. First, simple random sampling fixes the sample size, so every realized sample presents the same assignment problem: n units, of which the design sends exactly n_T to treatment. Second, the assignment mechanism is the same for every realized sample; the design does not, say, change the count of treated units according to which units it happens to draw. Under these two conditions, listing each sample's units in a fixed order (by superpopulation index, say) identifies every $\Omega_{\boldsymbol{r}}$ with the same set of binary n-vectors, and Ω denotes that common object. After the identification, the assignment vector is independent of the sampling vector every sample faces the same lottery although which *units of the superpopulation* receive treatment of course depends on both stages. When one formula mixes the stages, as in the products $R_i Z_i$ of the companion guides, the index i on Z_i runs over the superpopulation; for an unsampled unit the product is zero whatever value Z_i might take, so no definition of Z_i off the sample is ever needed. Under Bernoulli sampling the identification would fail at the first step: The length of the assignment vector itself would be random.

The potential outcomes deserve the same two-stage scrutiny, in the same numbered format as before, and the schedule itself needs a wider domain and a wider codomain before any assumption is invoked.

Step 1 (general object). With both stages in view, the schedule becomes a function on whole configurations,

$$\boldsymbol{y} : \Pi \times \Omega \to \mathbb{R}^N,$$

mapping each pair $(\boldsymbol{r}, \boldsymbol{z})$ a sample and an assignment of that sample to the N-vector of outcomes that *all* units of $\mathcal{P}_N$ would exhibit under that configuration, the unsampled units included: Their entries record the outcomes they would exhibit while standing outside the study. Unit i's coordinate is $y_i(\boldsymbol{r}, \boldsymbol{z})$, and the schedule of the earlier sections is the within-sample restriction: Fixing $\boldsymbol{r}$ and reading off the sampled coordinates recovers a function from $\{0,1\}^n$ to $\mathbb{R}^n$, one for each sample.

Step 2 (assumption invoked). A configuration gives unit i one of two statuses: *treated*, when the sample includes unit i and the assignment sends that unit to treatment, and *untreated* otherwise, whether the unit sits in the control arm or outside the sample entirely. I extend the stability assumption to both stages: $y_i(\boldsymbol{r}, \boldsymbol{z})$ depends on the configuration only through unit i's own status.

The extension says two things at once. First, no unit's outcome depends on which other units the design samples or on how the design assigns them. Second, being sampled does not itself change any outcome, so the sampled-control condition and the unsampled condition are outcome-equivalent. That last equivalence is substantive: It reads the control condition as the absence of treatment, and if control were instead an active intervention a placebo, an alternative program treating the two conditions as one would be an assumption in its own right.

Step 3 (licensed simplification). Every unit of $\mathcal{P}_N$ then has just two fixed potential outcomes, $y_i(1)$ and $y_i(0)$, defined whether or not the sampling stage ever selects the unit, with $y_i(0)$ doing double duty as the outcome under sampled control and the outcome outside the study. In the superpopulation sections the vectors $\boldsymbol{y}(1)$, $\boldsymbol{y}(0)$ and $\boldsymbol{\tau}$ accordingly collect N entries rather than n, and sample-level quantities recover the sampled entries through the indicators R_i. The estimand below is well defined precisely because unsampled units have potential outcomes too.

Step 4 (observability). For a sampled unit the observable outcome is $Y_i = y_i(Z_i)$, exactly as before; for an unsampled unit both potential outcomes exist, but neither is observable. Sampling and assignment never create or change potential outcomes; the two stages only determine which of the fixed values we see.

The estimand is the PATE,

$$\tau_{\text{PATE}} \coloneqq N^{-1} \sum_{i=1}^{N} \tau_i,$$

and the corresponding population variances, which follow the naming rule of the previous section (letter S, divisor $N-1$, subscript N for $\mathcal{P}_N$), are

$$\begin{aligned}
S_N^2(\boldsymbol{y}(1)) &= \left(\frac{1}{N-1}\right) \sum_{i=1}^{N} \left(y_i(1) - \frac{1}{N} \sum_{i=1}^{N} y_i(1) \right)^2 \\
S_N^2(\boldsymbol{y}(0)) &= \left(\frac{1}{N-1}\right) \sum_{i=1}^{N} \left(y_i(0) - \frac{1}{N} \sum_{i=1}^{N} y_i(0) \right)^2 \\
S_N^2(\boldsymbol{\tau}) &= \left(\frac{1}{N-1}\right) \sum_{i=1}^{N} \left(\tau_i - \tau_{\text{PATE}} \right)^2 .
\end{aligned}$$

The divisor $N-1$ pays off in the PATE variance guide: With this convention, the sample variance $S_n^2(\cdot)$ is *exactly* unbiased for $S_N^2(\cdot)$ under simple random sampling, with no correction factor.

In this setting the Difference-in-Means estimator has *two* sources of randomness, sampling $\{R_i\}_{i=1}^N$ and assignment $\{Z_i\}_{i=1}^n$. I write $\mathrm{E}_\Pi[\cdot]$ and $\mathrm{Var}_\Pi[\cdot]$ for expectations and variances over the sampling distribution, $\mathrm{E}_\Omega[\cdot]$ and $\mathrm{Var}_\Omega[\cdot]$ for those over the assignment distribution (conditional on the sample), and the unsubscripted E[·] and Var[·] for those over both. The law of total variance links the overall variance to the stage-specific moments,

$$\mathrm{Var}\left[\hat{\tau}\right] = \mathrm{E}_{\Pi}\left[\mathrm{Var}_{\Omega}\left[\hat{\tau}\right]\right] + \mathrm{Var}_{\Pi}\left[\mathrm{E}_{\Omega}\left[\hat{\tau}\right]\right],$$

which is the organizing identity for the PATE variance derivation.

One last convention covers the infinite superpopulation. As $N \to \infty$ with the population variances bounded, the quantities $S_N^2(\cdot)$ converge to limits that I write $\sigma^2(\cdot)$, with no subscript; these limiting superpopulation variances govern the infinite superpopulation results in the PATE variance guide.

References

Cox, D. R. (1958). *Planning of Experiments*. New York: John Wiley & Sons. 4

Freedman, D. A. (2009). *Statistical Models: Theory and Practice* (2nd ed.). New York: Cambridge University Press. 3

Gerber, A. S. and D. P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation*. New York: W. W. Norton. 3, 5

Pashley, N. E., G. W. Basse, and L. W. Miratrix (2021). Conditional as-if analyses in randomized experiments. *Journal of Causal Inference 9*(1), 264–284. 3

Rosenbaum, P. R. (2017). *Observation and Experiment: An Introduction to Causal Inference*. Cambridge, MA: Harvard University Press. 3

Rubin, D. B. (1980). Comment on "Randomization Analysis of Experimental Data: The Fisher Randomization Test" by D. Basu. *Journal of the American Statistical Association 75*(371), 591–593. 4

Rubin, D. B. (1986). Comment on "Statistics and Causal Inference" by P. W. Holland: Which ifs have causal answers. *Journal of the American Statistical Association 81*(396), 961–962. 4